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United States Patent Application Publication

20250281693

Kind Code

A1

Publication Date

September 11, 2025

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METHOD AND APPARATUS FOR THE FAIL-SAFE TERMINATION OF IN VIVO DRUG DELIVERY FROM AN IMPLANTABLE DRUG DELIVERY SYSTEM

Abstract

A method for the fail-safe termination of in vivo drug delivery from an implantable drug delivery system, the method comprising: providing an implantable drug delivery system comprising: a housing having a reservoir for containing a drug, and a port for dispensing the drug to a patient; and an emergency deactivation unit disposed between the reservoir and the port, the emergency deactivation unit comprising a composite structure comprising a biocompatible ferromagnetic mesh open to fluid flow and a hydrophobic meltable material, the hydrophobic meltable material comprising at least one hole therein for enabling a fluid to pass through the hydrophobic meltable material; implanting the implantable drug delivery system within a patient; enabling the drug to flow from the reservoir, through the at least one hole in the hydrophobic meltable material and out the port; and when drug flow is to be terminated, applying a magnetic field to the composite structure, such that a current is induced in the ferromagnetic mesh which heats the ferromagnetic mesh and melts the hydrophobic meltable material, thereby closing the at least one hole in the hydrophobic meltable material and blocking drug delivery to the patient.

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Family ID: 58558055

Appl. No.: 19/217073

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18388626 20231110 parent-grant-document US 12337141 child US 19217073

parent US continuation 17209919 20210323 parent-grant-document US 11813432 child US 18388626
parent US continuation 15769401 20180419 parent-grant-document US 10953154 US continuation PCT/US2016/057628 20161019 child US 17209919
us-provisional-application US 62243265 20151019

Publication Classification

Int. Cl.: **A61M5/168** (20060101); **A61K9/00** (20060101); **A61K47/02** (20060101); **A61K47/34** (20170101); **A61M5/14** (20060101); **A61M5/142** (20060101); **F16K13/10** (20060101); **F16K17/38** (20060101); **F16K99/00** (20060101)

U.S. Cl.:

CPC **A61M5/16881** (20130101); **A61K9/0009** (20130101); **A61K9/0024** (20130101); **A61K47/02** (20130101); **A61K47/34** (20130101); **A61M5/14276** (20130101); **F16K13/10** (20130101); **F16K17/38** (20130101); **F16K99/003** (20130101); **F16K99/0044** (20130101); A61M5/14 (20130101); A61M2205/02 (20130101); A61M2205/0288 (20130101); A61M2205/3523 (20130101); A61M2207/00 (20130101); F16K2099/0088 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/388,626, filed Nov. 10, 2023, which is a continuation of U.S. patent application Ser. No. 17/209,919, filed Mar. 23, 2021, now U.S. Pat. No. 11,813,432, issued Nov. 14, 2023, which is a continuation of U.S. patent application Ser. No. 15/769,401, filed Apr. 19, 2018, now U.S. Pat. No. 10,953,154, issued Mar. 23, 2021, which is a 371 U.S. National Stage of International Application No. PCT/US2016/057628, filed Oct. 19, 2016, which claims the benefit of U.S. Provisional Application No. 62/243,265, filed Oct. 19, 2015, each of which is hereby incorporated herein by reference in their entireties for all purposes.

FIELD OF THE INVENTION

[0002] This invention relates to implantable drug delivery systems in general, and more particularly to methods and apparatus for the fail-safe termination of in vivo drug delivery from an implantable drug delivery system.

BACKGROUND OF THE INVENTION

[0003] Implantable drug delivery systems are well known in the art. Such implantable drug delivery systems are highly advantageous inasmuch as they can continuously release a given therapeutic drug and eliminate issues with patient compliance to a prescribed regimen.

Furthermore, next-generation implantable drug delivery systems are enabling remote electronic control of drug delivery so as to provide telemedical treatment options to healthcare providers.

[0004] Unfortunately, current implantable drug delivery systems suffer from the disadvantage that, in the event that drug delivery must be terminated (e.g., due to an adverse patient reaction), and where the drug delivery system uses a passive release control (which is typical in the field) or where there is a malfunction of the electronic controller of the drug delivery system, etc., surgical explantation may be the only reliable option for terminating drug delivery. However, inasmuch as surgical explantation requires trained personnel, specialized and sterilized surgical equipment, and proper medical facilities, surgical explantation can be problematic, particularly in emergency

situations.

[0005] Thus, there is need for a new method and apparatus for the rapid and effective termination of in vivo drug delivery from an implantable drug delivery system. The new method and apparatus should, ideally, allow minimally-trained personnel, or the patients themselves, to safely halt drug delivery until appropriate surgical explantation can be effected. The new method and apparatus should also, ideally, be activatable from outside the body and be fail-safe so as to ensure reliable termination of drug delivery when activated. Additionally, the new method and apparatus should be readily compatible with a wide range of drug delivery systems so as to facilitate widespread adoption of the technology.

SUMMARY OF THE INVENTION

[0006] The present invention comprises the provision and use of an implantable drug delivery system which comprises a housing having a reservoir for containing a drug, and a port for dispensing the drug to the patient. In accordance with the present invention, the implantable drug delivery system further comprises an emergency deactivation unit which normally passes the drug therethrough but which, upon the application of energy to the emergency deactivation unit, prevents the drug from passing through the emergency deactivation unit (and hence out of the implantable drug delivery system). In one preferred form of the invention, the emergency deactivation unit comprises a composite structure comprising a biocompatible ferromagnetic mesh and a meltable material. The meltable material normally has one or more holes therein which allow the drug to pass from the reservoir, through the composite structure, and out the port of the implantable drug delivery system. However, when a magnetic field is applied to the composite structure, a current is induced in the ferromagnetic mesh which heats the ferromagnetic mesh and melts the meltable material, thereby closing the hole(s) in the composite structure and blocking drug delivery to the patient.

[0007] Significantly, the emergency deactivation unit may be used with a wide range of drug delivery systems so as to facilitate widespread adoption of the technology. Furthermore, the emergency deactivation unit may be used to close substantially any implantable fluid pathway, including implantable fluid pathways used for devices other than drug delivery systems.

[0008] In one preferred form of the present invention, there is provided a method for the fail-safe termination of in vivo drug delivery from an implantable drug delivery system, the method comprising: [0009] providing an implantable drug delivery system comprising: [0010] a housing having a reservoir for containing a drug, and a port for dispensing the drug to a patient; and [0011] an emergency deactivation unit disposed between the reservoir and the port, the emergency deactivation unit comprising a composite structure comprising a biocompatible ferromagnetic mesh open to fluid flow and a hydrophobic meltable material, the hydrophobic meltable material comprising at least one hole therein for enabling a fluid to pass through the hydrophobic meltable material; [0012] implanting the implantable drug delivery system within a patient; [0013] enabling the drug to flow from the reservoir, through the at least one hole in the hydrophobic meltable material and out the port; and [0014] when drug flow is to be terminated, applying a magnetic field to the composite structure, such that a current is induced in the ferromagnetic mesh which heats the ferromagnetic mesh and melts the hydrophobic meltable material, thereby closing the at least one hole in the hydrophobic meltable material and blocking drug delivery to the patient.

[0015] In another preferred form of the present invention, there is provided apparatus for the fail-safe termination of in vivo drug delivery from an implantable drug delivery system, the apparatus comprising: [0016] an implantable drug delivery system comprising: [0017] a housing having a reservoir for containing a drug, and a port for dispensing the drug to a patient; and [0018] an emergency deactivation unit disposed between the reservoir and the port, the emergency deactivation unit comprising a composite structure comprising a biocompatible ferromagnetic mesh open to fluid flow and a hydrophobic meltable material, the hydrophobic meltable material comprising at least one hole therein for enabling a fluid to pass through the hydrophobic meltable

material; [0019] wherein, when an appropriate magnetic field is applied to the composite structure, a current is induced in the ferromagnetic mesh which heats the ferromagnetic mesh and melts the hydrophobic meltable material, thereby closing the at least one hole in the hydrophobic meltable material and blocking drug delivery to the patient.

[0020] In another preferred form of the present invention, there is provided a method for the fail-safe termination of fluid flow through a fluid pathway, the method comprising: [0021] providing an emergency deactivation unit comprising a composite structure comprising a biocompatible ferromagnetic mesh open to fluid flow and a hydrophobic meltable material, the hydrophobic meltable material comprising at least one hole therein for enabling a fluid to pass through the hydrophobic meltable material; [0022] positioning the emergency deactivation unit within a fluid pathway; [0023] enabling fluid to flow from one side of the emergency deactivation unit, through the at least one hole in the hydrophobic meltable material, to the other side of the emergency deactivation unit; and [0024] when fluid flow is to be terminated, applying a magnetic field to the composite structure, such that a current is induced in the ferromagnetic mesh which heats the ferromagnetic mesh and melts the hydrophobic meltable material, thereby closing the at least one hole in the hydrophobic meltable material and blocking fluid flow through the emergency deactivation unit.

[0025] In another preferred form of the present invention, there is provided apparatus for the fail-safe termination of fluid flow through a fluid pathway, the apparatus comprising: [0026] an emergency deactivation unit comprising a composite structure comprising a biocompatible ferromagnetic mesh open to fluid flow and a hydrophobic meltable material, the hydrophobic meltable material comprising at least one hole therein for enabling a fluid to pass through the hydrophobic meltable material; [0027] wherein, when an appropriate magnetic field is applied to the composite structure, a current is induced in the ferromagnetic mesh which heats the ferromagnetic mesh and melts the hydrophobic meltable material, thereby closing the at least one hole in the hydrophobic meltable material and blocking fluid flow through the emergency deactivation unit.

[0028] In another preferred form of the present invention, there is provided a method for the fail-safe termination of in vivo drug delivery from an implantable drug delivery system, the method comprising: [0029] providing an implantable drug delivery system comprising: [0030] a housing having a reservoir for containing a drug, and a port for dispensing the drug to a patient; and [0031] an emergency deactivation unit disposed between the reservoir and the port, the emergency deactivation unit comprising: [0032] a barrier element comprising hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; and [0033] a magnetic heating element adapted to increase in temperature when exposed to a magnetic field, the magnetic heating element being disposed adjacent to the opening; [0034] implanting the implantable drug delivery system within a patient; [0035] enabling the drug to flow from the reservoir, through the opening in the barrier element and out the port; and [0036] when drug flow is to be terminated, applying a magnetic field to the magnetic heating element, such that the temperature of the magnetic heating element increases, causing the hydrophobic material of the barrier element to flow, whereby to close the opening in the barrier element and block drug delivery to the patient.

[0037] In another preferred form of the present invention, there is provided apparatus for the fail-safe termination of in vivo drug delivery from an implantable drug delivery system, the apparatus comprising: [0038] an implantable drug delivery system comprising: [0039] a housing having a reservoir for containing a drug, and a port for dispensing the drug to a patient; and [0040] an emergency deactivation unit disposed between the reservoir and the port, the emergency deactivation unit comprising: [0041] a barrier element comprising hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the

barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; and [0042] a magnetic heating element adapted to increase in temperature when exposed to a magnetic field, the magnetic heating element being disposed adjacent to the opening; [0043] wherein, when an appropriate magnetic field is applied to the magnetic heating element, the temperature of the magnetic heating element increases, causing the hydrophobic material of the barrier element to flow, whereby to close the opening in the barrier element and block drug delivery to the patient.

[0044] In another preferred form of the present invention, there is provided a method for the fail-safe termination of fluid flow through a fluid pathway, the method comprising: [0045] providing an emergency deactivation unit comprising: [0046] a barrier element comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; and [0047] a magnetic heating element adapted to increase in temperature when exposed to a magnetic field, the magnetic heating element being disposed adjacent to the opening; [0048] positioning the emergency deactivation unit within the fluid pathway; [0049] enabling fluid to flow from one side of the emergency deactivation unit, through the opening in the barrier element, to the other side of the emergency deactivation unit; and [0050] when fluid flow is to be terminated, applying a magnetic field to the magnetic heating element, such that the temperature of the magnetic heating element increases, causing the hydrophobic material of the barrier element to flow, whereby to close the opening in the barrier element and block fluid flow through the emergency deactivation unit.

[0051] In another preferred form of the present invention, there is provided apparatus for the fail-safe termination of fluid flow through a fluid pathway, the apparatus comprising: [0052] an emergency deactivation unit comprising: [0053] a barrier element comprising hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; and [0054] a magnetic heating element adapted to increase in temperature when exposed to a magnetic field, the magnetic heating element being disposed adjacent to the opening; [0055] wherein, when an appropriate magnetic field is applied to the magnetic heating element, the temperature of the magnetic heating element increases, causing the hydrophobic material of the barrier element to flow, whereby to close the opening in the barrier element and block fluid flow through the emergency deactivation unit.

[0056] In another preferred form of the present invention, there is provided a method for the fail-safe termination of fluid flow through a fluid pathway, the method comprising: [0057] providing an emergency deactivation unit comprising: [0058] a barrier element comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; [0059] positioning the emergency deactivation unit within the fluid pathway; [0060] enabling fluid to flow from one side of the emergency deactivation unit, through the opening in the barrier element, to the other side of the emergency deactivation unit; and [0061] when fluid flow is to be terminated, increasing the temperature of the barrier element, causing the hydrophobic material of the barrier element to flow, whereby to close the opening in the barrier element and block fluid flow through the emergency deactivation unit.

[0062] In another preferred form of the present invention, there is provided apparatus for the fail-safe termination of fluid flow through a fluid pathway, the apparatus comprising: [0063] an emergency deactivation unit comprising: [0064] a barrier element comprising hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; [0065] wherein appropriately increasing the temperature of the barrier element causes the hydrophobic material of the barrier element to flow, whereby to close the

opening in the barrier element and block fluid flow through the emergency deactivation unit. [0066] In another preferred form of the present invention, there is provided a method for manufacturing apparatus for the fail-safe termination of fluid flow through a fluid pathway, the method comprising: [0067] providing a substrate comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature; [0068] positioning a heating element on the substrate; [0069] positioning, atop the substrate and the heating element, a cover comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature; and [0070] adhering the cover to the substrate and forming an opening through the cover and the substrate adjacent to the heating element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0071] These and other objects and features of the present invention will be more fully disclosed or rendered obvious by the following detailed description of the preferred embodiments of the invention, which is to be considered together with the accompanying drawings wherein like numbers refer to like parts, and further wherein:

[0072] FIGS. 1A and 1B are schematic views showing an implantable drug delivery system with an emergency deactivation unit;

[0073] FIGS. 2A and 2B are schematic views showing the emergency deactivation unit incorporated in the implantable drug delivery system of FIGS. 1A and 1B;

[0074] FIG. 3 is a schematic view showing assembly of the emergency deactivation unit within the implantable drug delivery system;

[0075] FIG. 4 is a schematic view showing normalized release data demonstrating deactivation of drug release;

[0076] FIGS. 5A-5D are schematic views showing various dispositions of the emergency deactivation unit within the implantable drug delivery system;

[0077] FIGS. 6A-6D are schematic views showing various dispositions of an emergency deactivation unit within fluid passageways;

[0078] FIGS. 7A-7H are schematic views showing various configurations for the emergency deactivation unit; and

[0079] FIGS. 8A-8G are schematic views showing operation of various configurations of the emergency deactivation unit.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Implantable Drug Delivery System with Emergency Deactivation Unit

[0080] Looking first at FIGS. 1A and 1B, there is shown an implantable drug delivery system 5 (sometimes hereinafter referred to as the “nDS”) formed in accordance with the present invention. Implantable drug delivery system 5 generally comprises a housing 10 comprising a body 15 and a cap 20. Body 15 may be cylindrical and is closed off at one end by an end wall 25. A flange 30 is formed at the other end of body 15, opposite end wall 25. Flange 30 supports a membrane 35 (e.g., a silicon membrane or other rate-limiting membrane) which regulates the release of a drug contained in body 15 between end wall 25 and flange 30/membrane 35. Cap 20 may have a similar cross-sectional profile to that of body 15, and fits on body 15, so as to cover flange 30/membrane 35. Cap 20 comprises one or more holes 40. As a result of this construction, a drug contained within body 15 between end wall 25 and flange 30/membrane 35 may be selectively released by membrane 35 so as to thereafter pass out of housing 10 through hole(s) 40.

[0081] In accordance with the present invention, and looking next at FIGS. 1A, 1B, 2A and 2B, implantable drug delivery system 5 also comprises an emergency deactivation unit 45 for the fail-

safe termination of drug delivery from implantable drug delivery system 5. More particularly, emergency deactivation unit 45 generally comprises a composite structure 47 comprising a biocompatible ferromagnetic mesh 50 and a meltable material 55. Ferromagnetic mesh 50 comprises a biocompatible material which heats up when exposed to an appropriate magnetic field (i.e., a magnetic field below FDA Limits) due to the creation of a current within the ferromagnetic mesh. In one preferred form of the invention, ferromagnetic mesh 50 comprises stainless steel. Meltable material 55 may be any biocompatible material which (i) is solid within normal physiological temperature ranges (e.g., 30-45 degrees C.), and (ii) melts at a slightly higher temperature (e.g., 50-70 degrees C.). In one preferred embodiment of the invention, meltable material 55 comprises a soft polymer such as polycaprolactone (PCL).

[0082] The ferromagnetic mesh 50 is overlaid with (and preferably encapsulated by) meltable material 55, and meltable material 55 includes one or more holes 60, so that fluid (e.g., a drug) can normally pass through composite structure 47 of emergency deactivation unit 45 by passing through hole(s) 60 and ferromagnetic mesh 50. However, when composite structure 47 of emergency deactivation unit 45 is exposed to an appropriate magnetic field, the magnetic field induces a current in ferromagnetic mesh 50 which, according to the Joule effect, will increase the temperature of the ferromagnetic mesh. The heated ferromagnetic mesh 50 then melts meltable material 55, causing hole(s) 60 to close, whereby to prevent fluid (e.g., a drug) from passing through composite structure 47 of emergency deactivation unit 45.

[0083] In accordance with the present invention, composite structure 47 of emergency deactivation unit 45 is mounted within cap 20 so that the composite structure of the emergency deactivation unit resides between membrane 35 and hole(s) 40. Alternatively, composite structure 47 of emergency deactivation unit 45 may be positioned at another location between the drug reservoir and hole(s) 40. As a result of this construction, when implantable drug delivery system 5 is implanted within a patient with a drug contained within body 15 between end wall 25 and flange 30/membrane 35, membrane 35 may release the drug from body 15 so that the drug flows through hole(s) 60 of composite structure 47 of emergency deactivation unit 45, and then through hole(s) 40 of cap 20, so that the drug is administered to the patient. However, if it should become necessary for drug delivery to be terminated (e.g., due to an adverse patient reaction or malfunction of the drug delivery system, etc.), an appropriate magnetic field is introduced to composite structure 47 of emergency deactivation unit 45 so that the magnetic field causes ferromagnetic mesh 50 to heat up, thereby melting meltable material 55, causing hole(s) 60 to close and thereby terminating drug delivery by implantable drug delivery system 5.

[0084] Thus, it will be seen that the present invention provides for the rapid and effective termination of in vivo drug delivery from an implantable drug delivery system. The present invention allows minimally-trained personnel, or the patients themselves, to safely halt drug delivery until appropriate surgical explantation can be effected. The present invention is also activatable from outside the body and is fail-safe so as to ensure termination of drug delivery when activated. Additionally, the present invention is readily compatible with a wide range of drug delivery systems so as to facilitate widespread adoption of the technology.

Example

1. Construction

[0085] By way of example but not limitation, in one exemplary form of the present invention, emergency deactivation unit 45 comprises a composite structure 47 formed out of a series of layers, i.e., a stainless steel (SS) ferromagnetic mesh 50 layer inside a polycaprolactone (PCL) meltable material 55 layer.

[0086] The stainless steel ferromagnetic mesh 50 has a dimension of 10×10 mm, 150 μm thickness, and holes having a diameter of 300 μm. In this form of the invention, meltable material 55 is formed by solvent casting. More particularly, PCL (1 gr, 90 kDa) is dissolved in 5 ml of chloroform inside of a sealed bottle and agitated for 4 hours at room temperature (RT). After evaporating the

chloroform (12 hours, RT), the PCL layer is dried at RT in a vacuum for 12 hours. Experiments have shown that untreated PCL is hydrophobic and does not allow for drug passage through holes **60** in meltable material **55**. In order to increase the hydrophilicity of meltable material **55**, the meltable material (i.e., the PCL) may be treated with a 2N NaOH solution for 4 hours prior to vacuum drying for an additional 2 hours. Once dry, two circular pieces of meltable material **55** (having a diameter of 12 mm) can be cut from the PCL. One layer of meltable material **55** is melted through heating with a hotplate for 1 minute at 80° C., and the stainless steel ferromagnetic mesh **50** is pressed into the meltable material **55** (i.e., into the liquid polymer). Maintaining the hotplate at the same temperature, a second layer of meltable material **55** (i.e., a second PCL layer) is placed on the top of the stainless steel ferromagnetic mesh **50**. Compression during melting reduces the thickness of the final composite structure (made up of stainless steel ferromagnetic mesh **50** and meltable material **55**) to a polymeric-metallic “sandwich” (i.e., the composite structure of emergency deactivation unit **45**) having a thickness of about 700 μm. After cooling, five (5) through holes **60** (400 μm in diameter) are created symmetrically about the center of the meltable material **55**. The composite emergency deactivation unit **45** is placed inside a polyether ether ketone (PEEK) implantable capsule (i.e., body **15**) and adhered in place using a UV epoxy. A PEEK cap (i.e., cap **20**) is placed over the composite layer (i.e., over emergency deactivation unit **45**) at the top of the implantable capsule (i.e., at the top of body **25**, just beyond the location of flange **30**/membrane **35**).

2. Emergency Deactivation

[0087] In order to test implantable drug delivery system **5**, in one experiment, Rhodamine may be introduced in place of the drug contained in body **15** between end wall **25** and flange **30**/membrane **35** body **15**. After assembling the implantable drug delivery system **5**, and in order to test the functioning of emergency deactivation unit **45**, a custom spectroscopy cuvette having a volume of 13 ml was prepared. Implantable drug delivery system **5** was then placed perpendicular to the cuvette's cap and epoxied. A robotic, UV-Vis spectrophotometer was then used to collect absorbance measurements (545 nm) every minute for 16 hours. A stir bar (rotated at 1400 rpm) was placed at the bottom of the cuvette so as to maintain solution homogeneity. After 16 hours, implantable drug delivery system **5** was removed and placed in a 5 ml glass cuvette filled with 3 ml of water. The glass cuvette was placed within a magnetic field generator and a temperature sensor was placed inside the PEEK cap (i.e., inside cap **20**). In order to simulate the temperature of the body (i.e., normal physiological temperature), the resting temperature was set to 37° C. Once implantable drug delivery system **5** reached the resting temperature, the magnetic field generator was activated, applying a magnetic field of 12 kA/m.^{sup.2} for 1 minute and melting meltable material **55** (i.e., the PCL) of emergency deactivation unit **45** at about 50° C. After heating, the system was allowed to cool back to 37° C. A stereoscopic microscope was employed to inspect the emergency deactivation unit **45** (i.e., the PCL) and determine if the holes **60** were closed. After the RF (magnetic field) treatment, implantable drug delivery system **5** was replaced into the custom spectroscopy cuvette and the same solution as previously. UV-Vis absorbance measurements were taken every minute for another 16 hours. The release data (FIG. **4**) demonstrates a 90% drop in normalized drug release.

Generalization of Inventive Concept

[0088] It is possible to generalize the inventive concept of the present invention to include other embodiments.

1. Placement of the Emergency Deactivation Unit within an Implantable Drug Delivery System or within Devices Other than Implantable Drug Delivery Systems

[0089] The emergency deactivation unit may be placed in a variety of locations within an implantable drug delivery system or within devices other than implantable drug delivery systems.

[0090] By way of example but not limitation, and looking now at FIGS. **5A-5D**, where emergency deactivation unit **45** is placed within an implantable drug delivery system **5**, the emergency

deactivation unit **45** may be placed: [0091] (i) on the outside of a release-limiting membrane (e.g., on the outside of membrane **35**); or [0092] (ii) within the reservoir immediately before the membrane (e.g., between membrane **35** and end wall **25**); or [0093] (iii) directly on or against the membrane (e.g., on either side of membrane **35**); or [0094] (iv) sputtered/deposited onto the membrane itself (e.g., deposited directly on membrane **35**).

[0095] By way of further example but not limitation, and looking now at FIGS. **6A-6D**, where emergency deactivation unit **45** is placed within devices other than implantable drug delivery systems, the emergency deactivation unit **45** may be placed at the following locations in a general fluid pathway: [0096] (i) at the beginning of the fluid pathway; or [0097] (ii) at the end of the fluid pathway; or [0098] (iii) closing off a branching section of the fluid pathway.

[0099] Note that emergency deactivation unit **45** may be used to make a general fluid pathway more or less fluid tight, may receive additional heating to re-open one or more of the holes **60** in composite structure **47**, and may be employed in pressurized or non-pressurized fluid pathways.

2. Composition of the Meltable Material

[0100] The meltable material (e.g., meltable material **55**) may comprise any material which is solid at physiological temperatures and has a melting point somewhat higher than physiological temperatures (e.g., ≥ 5 degrees higher). By way of example but not limitation, the meltable material may comprise: [0101] (i) a polymer; or [0102] (ii) wax; or [0103] (iii) a shape memory alloy.

[0104] Note that the meltable material may be biodegradable or non-biodegradable, bioinert or non-bioinert.

[0105] Note also that the meltable material can be a singular material or a composite of meltable materials. Where the meltable material is a composite of meltable materials, the component materials can have the same melting point or can have differing melting points (e.g., to effect greater control over flow).

[0106] It should also be appreciated that the meltable material could be replaced by material which does not require melting, but instead uses other means (e.g., swelling) of closing the through holes (e.g., through holes **60**).

3. Magnetic Heating Material

[0107] The magnetic heating material (e.g., ferromagnetic mesh **50**) may be provided in a variety of configurations. By way of example but not limitation, and looking now at FIGS. **7A-7H**, the magnetic heating material may be: [0108] (i) a mesh of different sizes; or [0109] (ii) a mesh limited to immediately around the through holes (e.g., through holes **60**); or [0110] (iii) a single ring or multiple rings, either concentric or not; or [0111] (iv) any geometric shape allowing melt-based closure of the through holes; or [0112] (v) a single sheet within the meltable material; or [0113] (vi) a deposited layer on the meltable material (e.g., on top of, or below, or both).

[0114] Note that the magnetic heating material may also comprise ferromagnetic powder/shavings distributed throughout the meltable material.

[0115] The magnetic heating material (e.g., ferromagnetic mesh **50**) may be formed out of a variety of materials. By way of example but not limitation, the magnetic heating material may comprise:

[0116] (i) a ferromagnetic metal; or [0117] (ii) a paramagnetic material; or [0118] (iii) any material that will heat up in a magnetic field.

4. How Fluid Passage is Stopped

[0119] The emergency deactivation unit of the present invention may stop fluid flow in a variety of ways. By way of example but not limitation, and looking now at FIGS. **8A-8G**, the emergency deactivation unit may stop fluid flow by: [0120] (i) closure of through holes **60** formed in composite structure **47**; or [0121] (ii) sealing of composite structure **47** against fluid pathway surfaces; or [0122] (iii) running meltable material along a directed route to stop fluid flow through a pathway (either singular or branching)—the meltable material may be run along the directed route in a variety of ways (e.g., along a guide such as a mesh or struts, using gravity, utilizing layer placement, utilizing chemical properties, utilizing a pressure gradient in fluid, etc.).

[0123] Note that the through holes to be closed (e.g., through holes **60**) can be in the same layer as the meltable material or can be in a secondary layer (into which the meltable material flows).

[0124] Note also that heating of the meltable material can be used to effect complete stopping of fluid flow or partial stopping of fluid flow.

Modifications of the Preferred Embodiments

[0125] It should be understood that many additional changes in the details, materials, steps and arrangements of parts, which have been herein described and illustrated in order to explain the nature of the present invention, may be made by those skilled in the art while still remaining within the principles and scope of the invention.

Claims

1. A method for the fail-safe termination of fluid flow through a fluid pathway, the method comprising: providing an emergency deactivation unit comprising: a barrier element comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; positioning the emergency deactivation unit within the fluid pathway; enabling fluid to flow from one side of the emergency deactivation unit, through the opening in the barrier element, to the other side of the emergency deactivation unit; and when fluid flow is to be terminated, increasing the temperature of the barrier element, causing the hydrophobic material of the barrier element to flow, whereby to close the opening in the barrier element and block fluid flow through the emergency deactivation unit.
 2. Apparatus for the fail-safe termination of fluid flow through a fluid pathway, the apparatus comprising: an emergency deactivation unit comprising: a barrier element comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature, the barrier element comprising an opening extending therethrough for allowing fluid to pass through the barrier element; wherein appropriately increasing the temperature of the barrier element causes the hydrophobic material of the barrier element to flow, whereby to close the opening in the barrier element and block fluid flow through the emergency deactivation unit.
 3. A method for manufacturing apparatus for the fail-safe termination of fluid flow through a fluid pathway, the method comprising: providing a substrate comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature; positioning a heating element on the substrate; positioning, atop the substrate and the heating element, a cover comprising a hydrophobic material having a solid state at physiological temperature and a flowable state above physiological temperature; and adhering the cover to the substrate and forming an opening through the cover and the substrate adjacent to the heating element.
 4. A method according to claim 3, wherein the hydrophobic material of the substrate, and the hydrophobic material of the cover, both comprise polycaprolactone.
 5. A method according to claim 3, wherein the heating element comprises a magnetic heating element adapted to increase in temperature when exposed to a magnetic field.
 6. A method according to claim 5, wherein the magnetic heating element comprises a ferromagnetic mesh.
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